

Cohort Definition Examples

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```
library(Capr)

# Function to test that cohort compiles
assertCohortCompiles <- function(cohortDefinition) {
  sql <- CirceR::cohortExpressionFromJson(as.json(ch))
  sql <- CirceR::buildCohortQuery(sql, options = CirceR::createGenerateOptions(generateStats = TRUE))

  stopifnot(nchar(sql) > 1)
  invisible(NULL)
}
```

This vignette provides a number of example cohorts built with Capr from the OHDSI *Phenotype Phebruary* event.

1 Type 2 diabetes mellitus

Persons with new type 2 diabetes mellitus at first diagnosis

<https://atlas-phenotype.ohdsi.org/#/cohortdefinition/88/definition>

```
library(Capr)

cs0 <- cs(descendants(443238, 201820, 442793),
          descendants(exclude(195771, 201254, 435216, 761051, 4058243, 40484648)),
          name = "Type 2 diabetes mellitus (diabetes mellitus excluding T1DM and secondary)")

ch <- cohort(
  entry = entry(
    conditionOccurrence(cs0),
    observationWindow = continuousObservation(priorDays = 365)
  ),
  exit = exit(
    endStrategy = observationExit()
  )
)
```

Persons with new type 2 diabetes and no prior T1DM or secondary diabetes

<https://atlas-phenotype.ohdsi.org/#/cohortdefinition/89/export>

```
library(Capr)

cs0 <- cs(descendants(443238, 201820, 442793),
          descendants(exclude(195771, 201254, 435216, 761051, 4058243, 40484648)),
          name = "Type 2 diabetes mellitus (diabetes mellitus excluding T1DM and secondary)")

cs1 <- cs(descendants(201254, 435216, 40484648),
          name = "Type 1 diabetes mellitus")

cs2 <- cs(descendants(195771),
          name = "Secondary diabetes mellitus")

ch <- cohort(
  entry = entry(
    conditionOccurrence(cs0),
    observationWindow = continuousObservation(priorDays = 365)
  ),
  attrition = attrition(
    't1d' = withAll(
      exactly(0, conditionOccurrence(cs1), duringInterval(eventStarts(-Inf, 0)))
    ),
    'secondaryDiabetes' = withAll(
      exactly(0, conditionOccurrence(cs2), duringInterval(eventStarts(-Inf, 0)))
    )
  ),
  exit = exit(
    endStrategy = observationExit()
  )
)
```

Persons with new type 2 diabetes mellitus at first dx rx or lab

<https://atlas-phenotype.ohdsi.org/#/cohortdefinition/90>

```
library(Capr)

cs0 <- cs(descendants(443238, 201820, 442793),
          descendants(exclude(195771, 201254, 435216, 761051, 4058243, 40484648)),
          name = "Type 2 diabetes mellitus (diabetes mellitus excluding T1DM and secondary)")

cs1 <- cs(descendants(201254, 435216, 40484648),
          name = "Type 1 diabetes mellitus")

cs2 <- cs(descendants(195771),
          name = "Secondary diabetes mellitus")

cs3 <- cs(descendants(4184637, 37059902),
          name = "Hemoglobin A1c (HbA1c) measurements")

cs4 <- cs(descendants(21600744),
          name = "Drugs for diabetes except insulin")

ch <- cohort(
```

```

entry = entry(
  conditionOccurrence(cs0),
  drugExposure(cs4),
  measurement(cs3, valueAsNumber(bt(6.5, 30)), measurementUnit("%")),
  measurement(cs3, valueAsNumber(bt(48, 99)), measurementUnit("mmol/mol")),
  observationWindow = continuousObservation(priorDays = 365)
),
attrition = attrition(
  'no T1D' = withAll(
    exactly(0, conditionOccurrence(cs1), duringInterval(eventStarts(-Inf, 0)))
  ),
  'no secondary diabetes' = withAll(
    exactly(0, conditionOccurrence(cs2), duringInterval(eventStarts(-Inf, 0)))
  )
),
exit = exit(
  endStrategy = observationExit()
)
)

```

2 Type 1 diabetes mellitus

Persons with new type 1 diabetes

<https://atlas-phenotype.ohdsi.org/#!/cohortdefinition/92>

```

cs0 <- cs(descendants(195771),
  name = "Type 1 diabetes mellitus")

ch <- cohort(
  entry = entry(
    conditionOccurrence(cs0),
    observationWindow = continuousObservation(priorDays = 365)
  )
)

```

```
assertCohortCompiles(ch)
```

Persons with new type 1 diabetes and no prior T2DM or secondary diabetes

<https://atlas-phenotype.ohdsi.org/#!/cohortdefinition/91>

```

cs0 <- cs(descendants(443238, 201820, 442793),
  descendants(exclude(195771, 201254, 435216, 761051, 4058243, 40484648)),
  name = "Type 2 diabetes mellitus (diabetes mellitus excluding T1DM and secondary)")

cs1 <- cs(descendants(201254, 435216, 40484648),
  name = "Type 1 diabetes mellitus")

cs2 <- cs(descendants(195771),
  name = "Secondary diabetes mellitus")

ch <- cohort(
  entry = entry(
    conditionOccurrence(cs1),

```

```

    observationWindow = continuousObservation(priorDays = 365)
  ),
  attrition = attrition(
    "no prior T2DM" = withAll(exactly(0, conditionOccurrence(cs0), duringInterval(eventStarts(-Inf, 0))),
    "no prior secondary T1DM" = withAll(exactly(0, conditionOccurrence(cs2), duringInterval(eventStarts
  )
)
)

```

3 Atrial Fibrillation

Persons with atrial fibrillation per Wharton et al 2021

<https://atlas-phenotype.ohdsi.org/#!/cohortdefinition/93>

```

cs0 <- cs(descendants(313217),
  name = "Atrial fibrillation")

ch <- cohort(conditionOccurrence(cs0))

```

Persons with atrial fibrillation per Subramanya et al 2021

<https://atlas-phenotype.ohdsi.org/#!/cohortdefinition/94>

```

afib <- cs(descendants(313217),
  name = "Atrial fibrillation")

ip <- cs(descendants(262, 9201),
  name = "Inpatient or inpatient ER visit")

op <- cs(descendants(9202, 9203),
  name = "Outpatient or ER visit")

ch <- cohort(
  entry = entry(
    conditionOccurrence(
      conceptSet = afib,
      nestedWithAll(
        # an inpatient afib
        atLeast(
          x = 1,
          aperture = duringInterval(eventStarts(-Inf, 0), eventEnds(0, Inf)),
          query = visit(conceptSet = ip)
        )
      )
    ),
  conditionOccurrence(
    conceptSet = afib,
    nestedWithAll(
      # 2 afib in outpatient
      atLeast(
        x = 1,
        aperture = duringInterval(eventStarts(-Inf, 0), eventEnds(0, Inf)),
        query = visit(
          conceptSet = op,

```

```
nestedWithAll(  
    atLeast(  
        x = 1,  
        aperture = duringInterval(eventStarts(7, 365)),  
        query = conditionOccurrence(  
            conceptSet = afib,  
            nestedWithAll(  
                atLeast(  
                    x = 1,  
                    aperture = duringInterval(eventStarts(-Inf, 0), eventEnds(0, Inf)),  
                    query = visit(conceptSet = op)  
                )  
            )  
        )  
    )  
)
```

4 Source Concept Example

```
lf <- cs(descendants(4197819), name = "lung fibrosis")
lfSource <- cs(
  725357,35208074,45543276,45548125,45552899,45562465,45567273,45581864,45586679,45586680,45601135,
  name = "Lung Fibrosis Source"
)

# Want cohort where entry is lung fibrosis but only using source ICD10CM codes

cd <- cohort(
  entry = entry(
    conditionOccurrence(
      conceptSet = lf,
      conditionSourceConcept(lfSource) # add source concept as attribute
    ),
    observationWindow = continuousObservation(365L, 0L),
    primaryCriteriaLimit = "First"
  )
)
```